

Sampling What We Cannot Sum

Markov Chain Monte Carlo at the Frontier

Lecture 4's **walker**, a Markov chain, visits each candidate answer in proportion to its probability, **without the impossible sum** Z . It is behind black-hole measurements, image generators and a new way to sample language models. Today you see it at the frontier, and try it yourself.

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Builds on Lecture 4: Sampling replaces an impossible sum

Outline

► Sampling without Z (Lecture 4)

Troubles 1–3: measuring unknowns from data

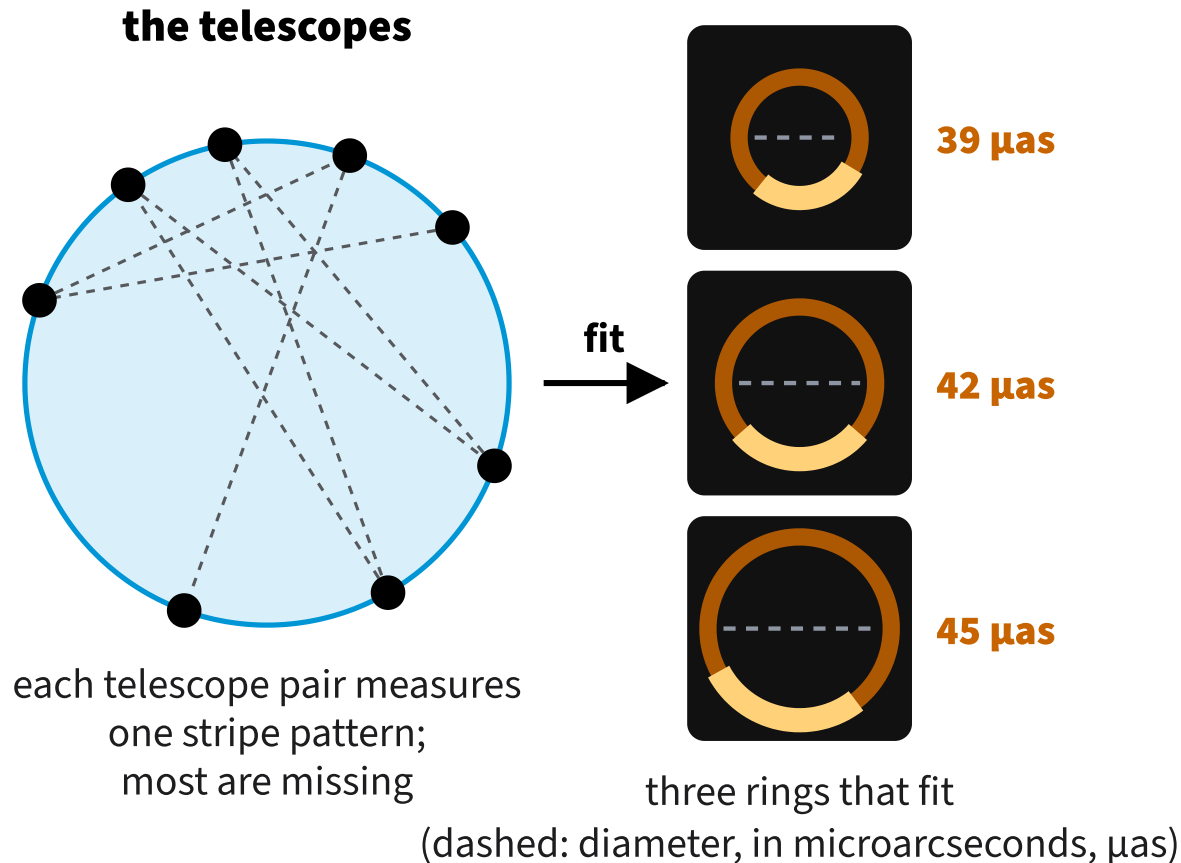
Troubles 4–5: creating pictures and text

Conclusion and your turn

One story: the sampler of Lecture 4, which we call **the walker**, meets five troubles at the frontier, and one small change fixes each. First: why sample at all?

Sampling without Z

April 2019: the first picture of a black hole.



Look: telescopes across the Earth gave only a few measurements, and all three rings on the left fit them.

Q: Which picture is right?

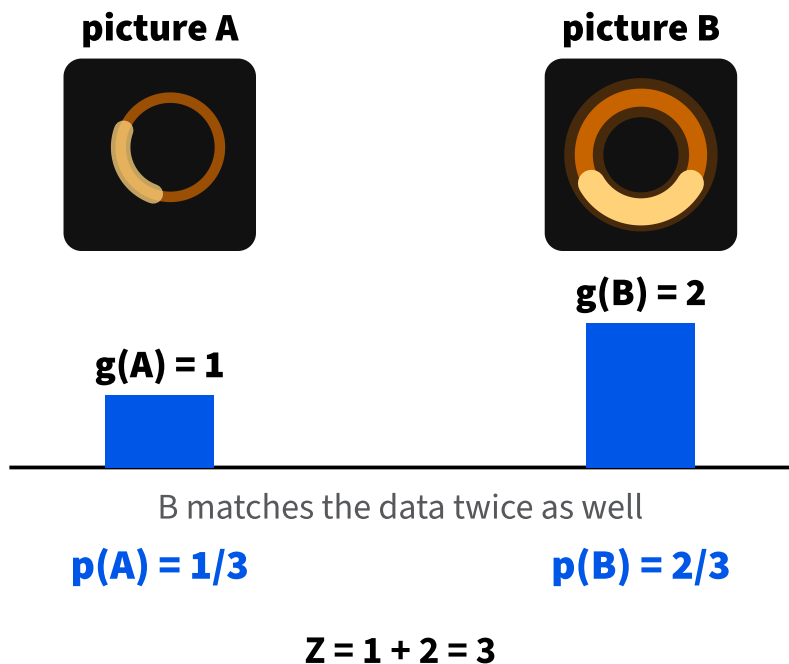
A: Many fit. Pick among them at random, better fits more often: their spread is the error bar.

Sampling without Z : in three sentences

- BACKGROUND** Data rarely pin down one answer: many answers fit, some better than others.
- KEY IDEA** Pick answers at random, better fits more often. Lecture 4's **walker** does this by comparing just two answers at a time, so it never needs the **impossible** total Z .
- USED FOR** Answers with error bars, from the size of a black hole to the effect of a medicine.

Sampling without Z

Main idea: sample, that is, pick a picture at random, with probability proportional to its **fit score**.



$$p(x) = \frac{g(x)}{Z}$$

← fit score: easy

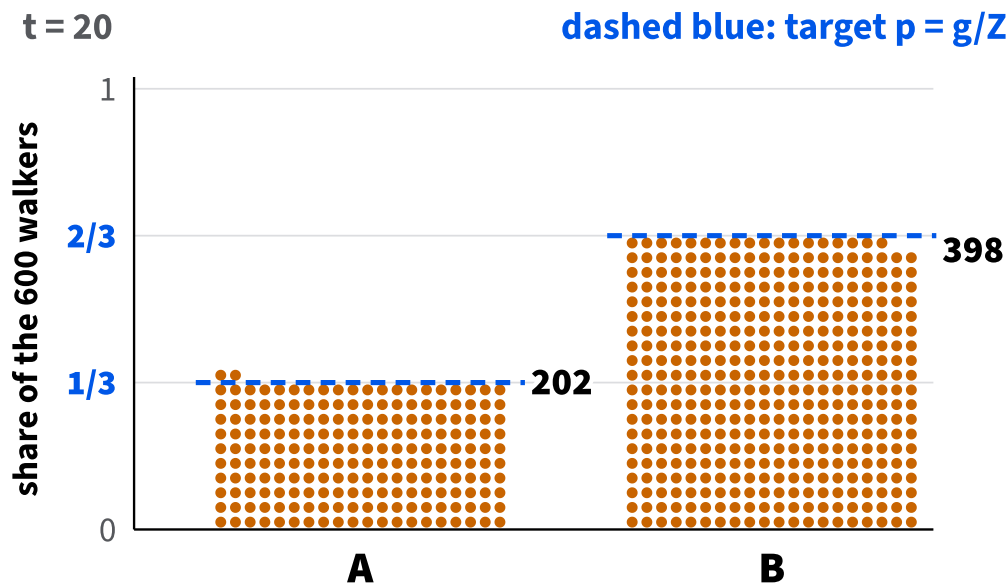
← sum of all fit scores: hard

Example Fit scores 1 and 2, so pick A one third and B two thirds of the time.

Look: one fit score is easy to compute. Here $Z = 1 + 2 = 3$, but for a real image the sum has more terms than there are atoms in the universe.

Sampling without Z

The trick: compare two pictures, and Z cancels.



$$\frac{p(\text{new})}{p(\text{old})} = \frac{g(\text{new})/Z}{g(\text{old})/Z} = \frac{g(\text{new})}{g(\text{old})}$$

The walker (a **Markov chain**) holds one picture. Each step it proposes another, and **the accept test** decides: move or stay [Metropolis et al., 1953; Hastings, 1970].

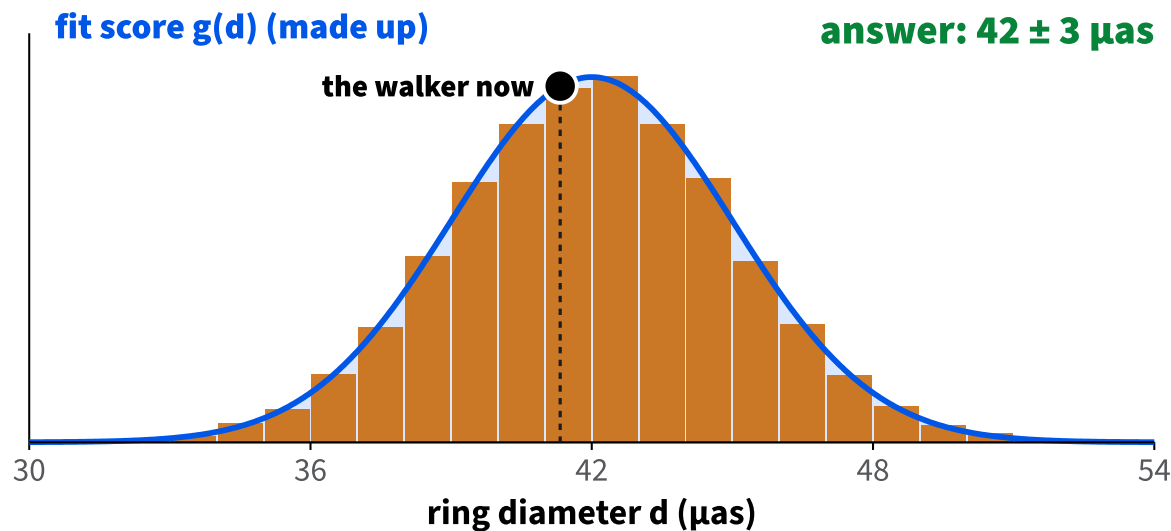
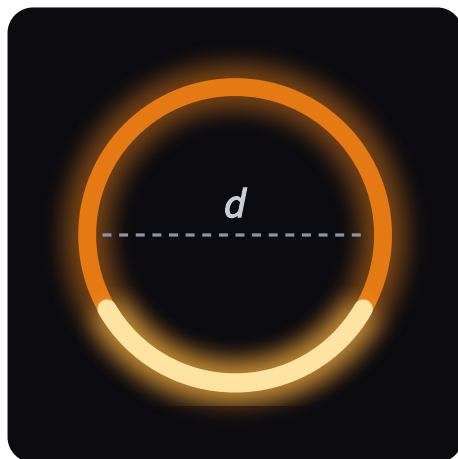
- ▶ **Better fit** ($A \rightarrow B$): always move.
- ▶ **Worse fit** ($B \rightarrow A$): move with probability $g(\text{new})/g(\text{old}) = \frac{1}{2}$.

Watch: 600 **walkers** start at A. Their **visits**, the places they have been, settle near one third at A and two thirds at B (**p. 35**). These visits are the **samples**.

Try: set the **fit score** of B to 3. Where will they settle?

Sampling without Z

One **walker** hops between ring sizes; where it spends its time gives **the answer**.



Watch: each step proposes a nearby size and applies the **accept test** (p. 6); the ring shows the current size, and the orange bars count the **visits**. Their centre is the answer, their width the **error bar**. **Try:** tiny or huge steps.

Q: Why not just take the best-fitting ring? A: It has no error bar, and nearby rings fit almost as well.

Result: 42 ± 3 microarcseconds (μas), about an orange on the Moon seen from Earth; mass 6.5 ± 0.7 billion Suns [Event Horizon Telescope Collaboration, 2019].

Main Question

**The walker needs only ratios of fit scores, never Z .
Can it also weigh colliding black holes, measure how
much COVID-19 lockdowns helped, give neural
networks error bars, paint pictures, and help language
models reason?**

This is indeed the case, after a few small changes.

Keep this question in mind: each part changes the same walker in one small way.

Sampling without Z

Six words for the rest of the lab.

Word	What it means today	Name in Lecture 4
candidate x	one possible answer: a picture, a ring size, a set of network weights, a sentence	a state
fit score $g(x)$	how well x matches the data; Z is the sum of all fit scores, and $p = g/Z$	weight (energy $E = -\ln g$)
the walker	one point that jumps between candidates: propose, then the accept test	Markov chain (Metropolis–Hastings)
accept test	keep a proposed jump with probability $\min\{1, g(\text{new})/g(\text{old})\}$ (p. 6)	acceptance step
visits	the candidates the walker has been at; their histogram approaches p	samples
hill, valley	a region of good fits; the bad fits between them. Lecture 4 drew the energy $-\ln g$, so its valleys are our hills.	mode, barrier

Read: so “the **visits** fill a hill” means “the **samples** cover a region of good answers”.

Sampling without Z

Five troubles at the frontier, five fixes.

	Trouble	Fix	Used for
1	far-apart hills	???	colliding black holes
2	many unknowns at once	???	COVID-19 lockdowns
3	too much data	???	neural-network error bars
4	no formula for g , only examples	???	image generators
5	sharpening: a new Z over all texts	???	language-model reasoning

Read: troubles 1–3 appear when the **walker** measures unknowns, 4–5 when it creates. The fixes are revealed one by one.

Outline

Sampling without Z (Lecture 4)

► **Troubles 1–3: measuring unknowns from data**

Troubles 4–5: creating pictures and text

Conclusion and your turn

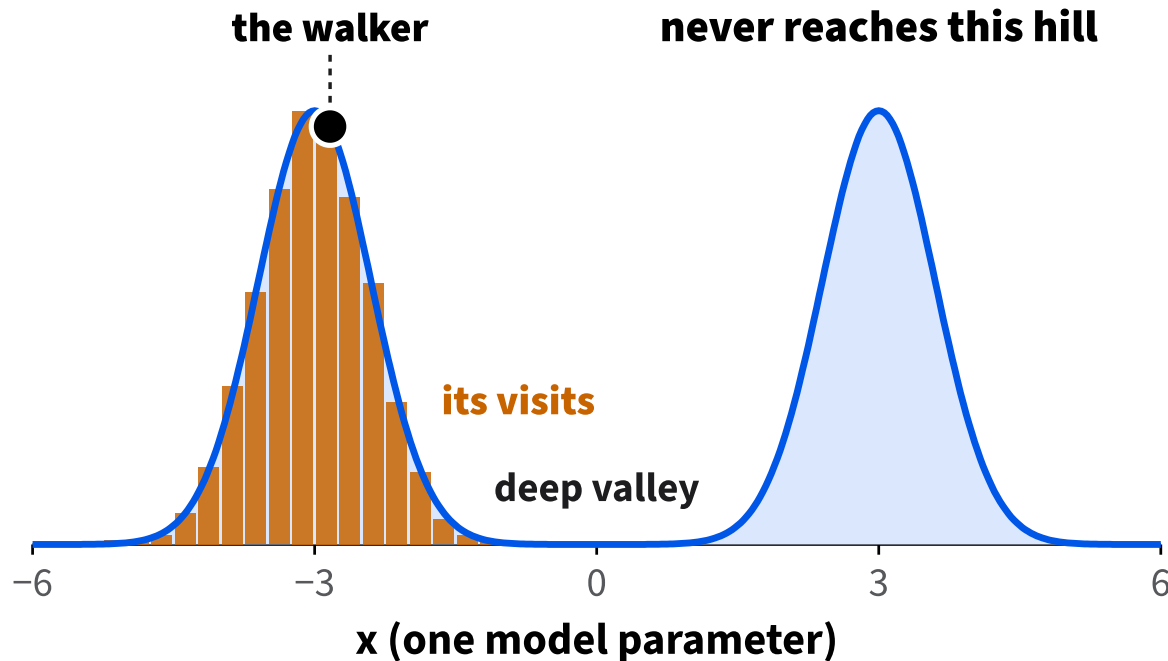
Next: three ways the walker fails when it measures, and the small change that fixes each.

Trouble 1 in three sentences

- BACKGROUND** Sometimes two very different answers both fit well (**two hills**), with bad answers in between.
- KEY IDEA** Run a second **walker** on **flattened fit scores**, so it crosses easily, and let the two **walkers swap** places now and then.
- USED FOR** Weighing two colliding black holes from a gravitational wave (LIGO, 2015), a discovery that won the 2017 Nobel Prize in Physics.

Trouble 1: far-apart hills

Trouble: two far-apart hills trap the walker.



Watch: its **visits** fill only the left hill. The right hill, an equally good group of answers, is **never found**, and nothing warns us.

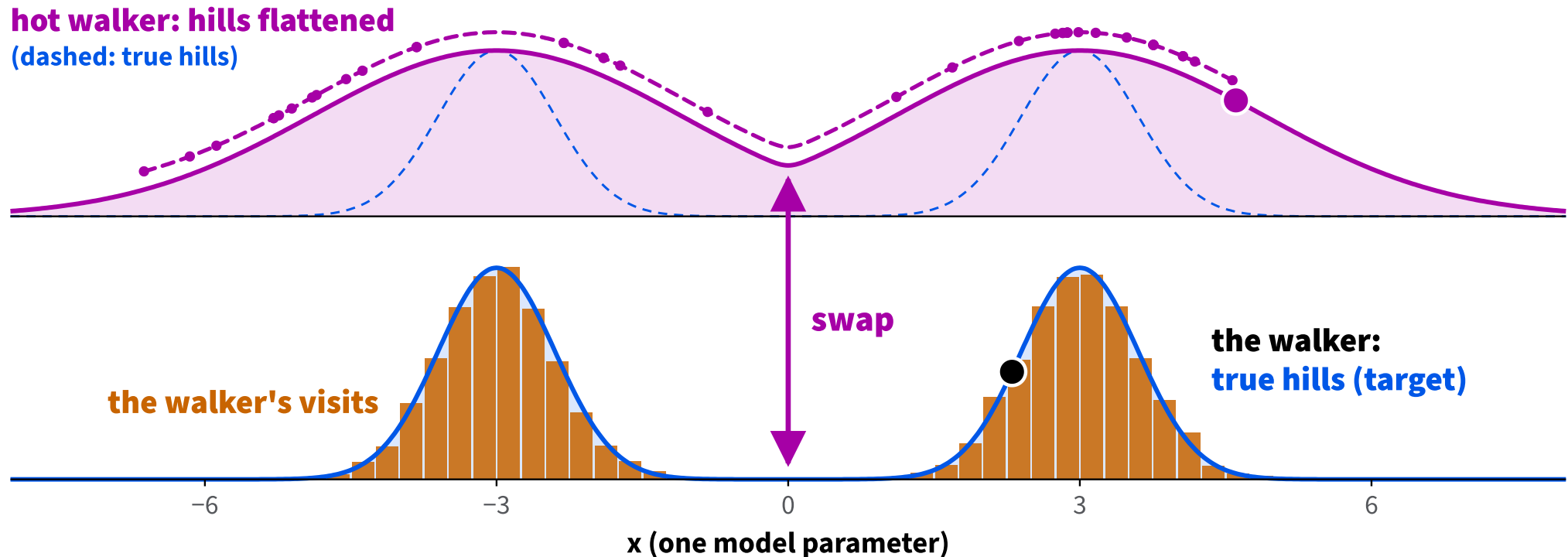
Q: Why is it stuck?

A: Crossing needs many worse-fit moves in a row, each rarely accepted.

Try: bring the hills closer.

Trouble 1: far-apart hills

Fix: run a second, **hot walker** on **flattened** hills; now and then the two **swap** places.



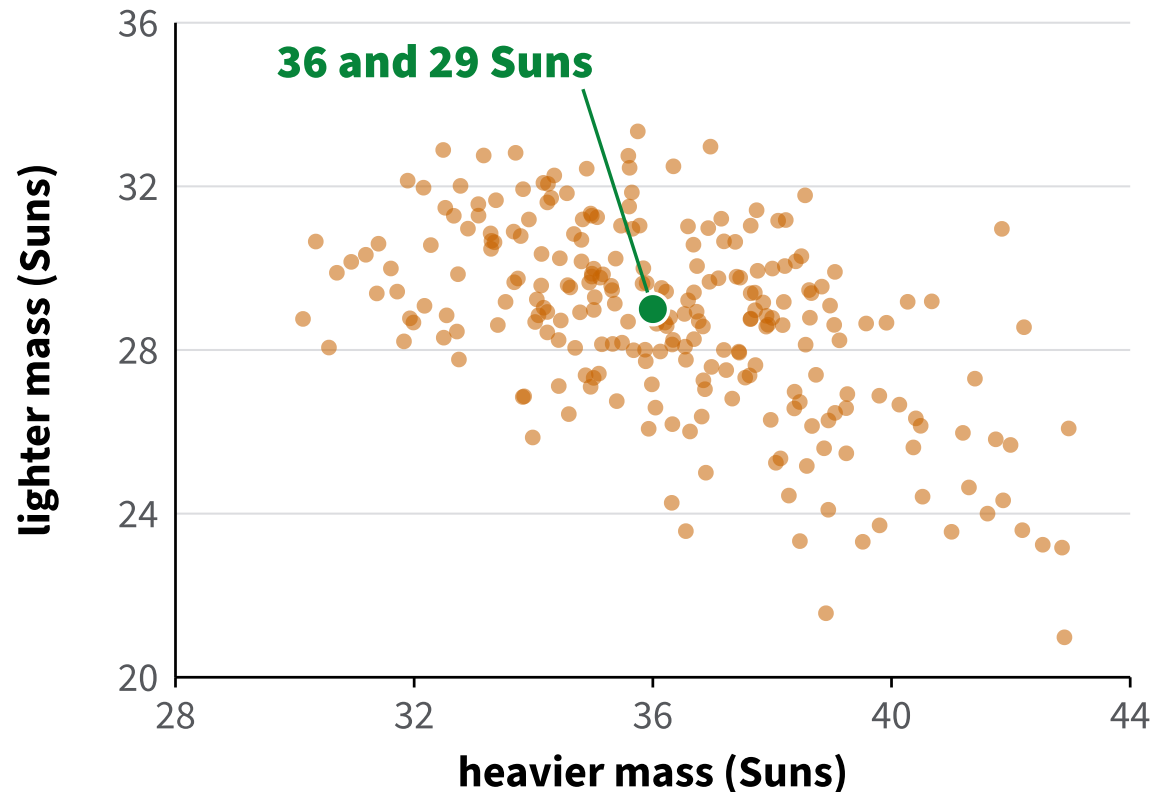
Watch: the hot walker (top) uses the **fit score** to the power $1/T$, which flattens the hills, so it crosses. At each magenta swap the two exchange positions after their own **accept test** (p. 37), so the walker's **visits** fill both hills.

Try: $T = 1$. Does the hot walker still help? This is **parallel tempering** [Swendsen and Wang, 1986; Geyer, 1991].

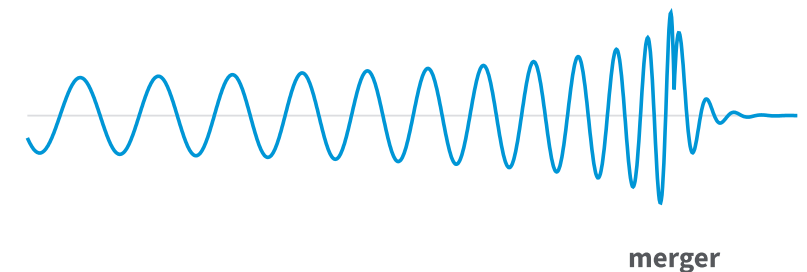
Trouble 1: far-apart hills

Result: two colliding black holes were weighed with hot and cold **walkers**.

each dot: one (illustrative) sample



the measured wave (schematic)



14 September 2015: the Laser Interferometer Gravitational-Wave Observatory (LIGO) caught a 0.2-second chirp, a rising whistle, from two black holes merging 1.3 billion light-years away [Veitch et al., 2015; Abbott et al., 2016]. It won the 2017 Nobel Prize in Physics.

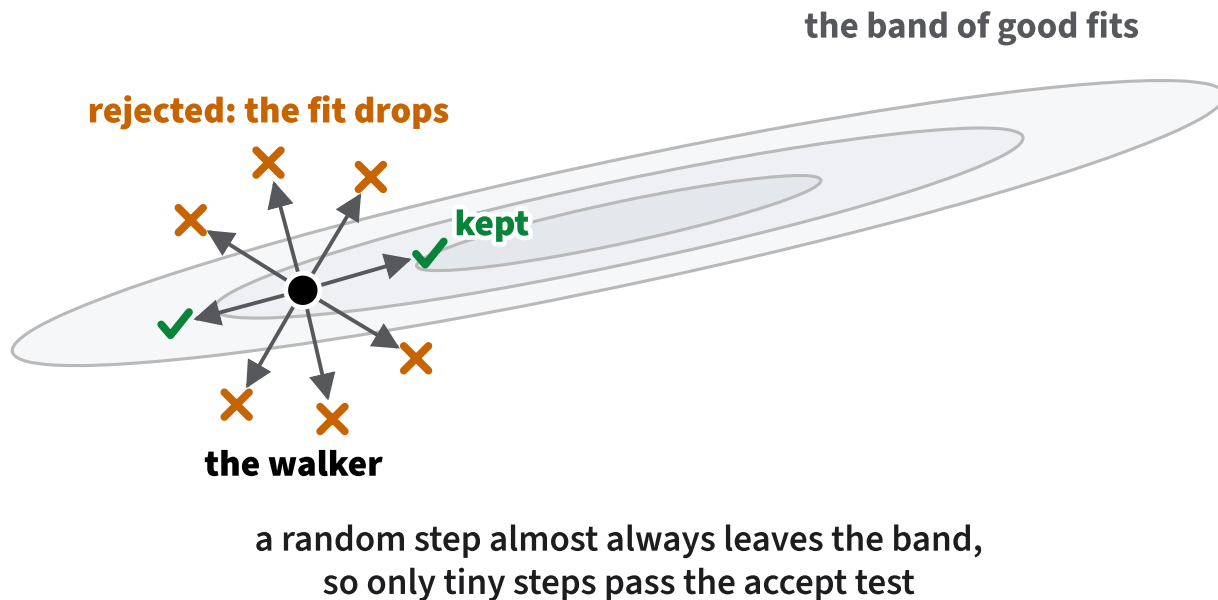
Look: each dot is one visit (a **sample**) of the two masses; the cloud's centre is the estimate, its size the **error bar**.

Trouble 2 in three sentences

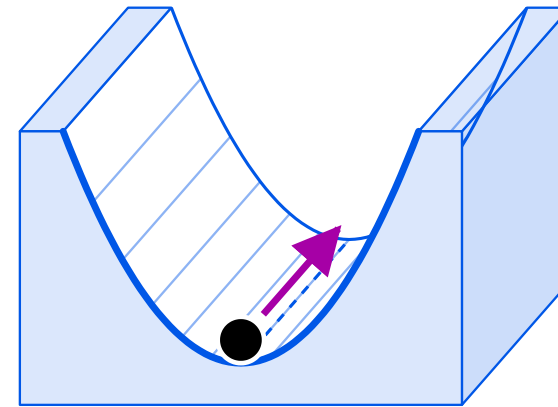
- BACKGROUND** Real models have hundreds of unknowns, and then a random step almost always lands on a bad answer.
- KEY IDEA** Give the **walker** a push and let it **slide like a marble** along the good answers, steered by the **slope** of the **fit score**.
- USED FOR** The Stan software, used to measure how much COVID-19 lockdowns helped in 11 countries (2020).

Trouble 2: many unknowns at once

Trouble: with many unknowns, good fits form a **narrow band**, and random steps leave it.



flip the fit upside down:
good fits = bottom of a groove
blue: the height $-\ln g$ across the band

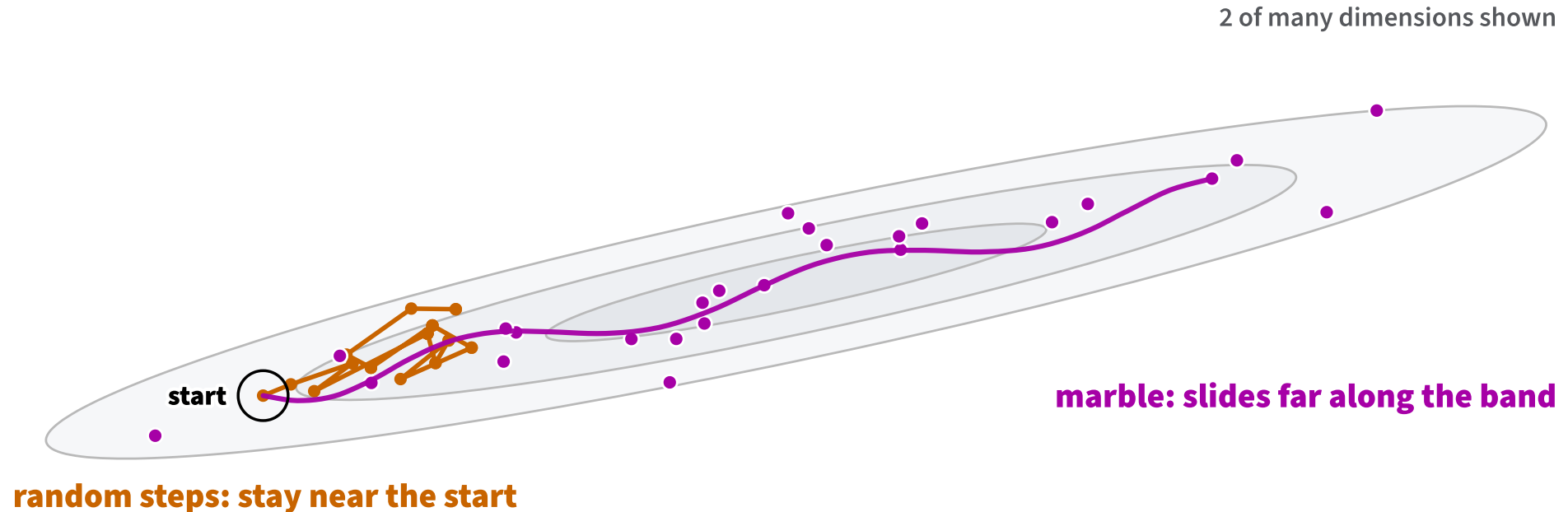


a pushed marble **slides along the groove**,
not out of it

Look: almost every random step leaves the band (X) and is rejected; only tiny steps are kept, so the **walker crawls**. On the right, the **fit score** is flipped upside down: the band becomes the bottom of a **groove**.

Trouble 2: many unknowns at once

Fix: push the **walker** and let it slide along the **groove**, like a **marble**.



grey: contour lines of the target · dots: where each roll ended

Watch: in 25 moves the random walker (orange) stays near the start, while the marble (magenta) slides far along the band. It keeps its energy, height plus $\text{speed}^2/2$, so even far moves pass the **accept test**. **Try:** roll length 1.

This is **Hamiltonian Monte Carlo** [Duane et al., 1987]; the Stan software used it to estimate how much lockdowns slowed COVID-19 in 11 European countries [Flaxman et al., 2020].

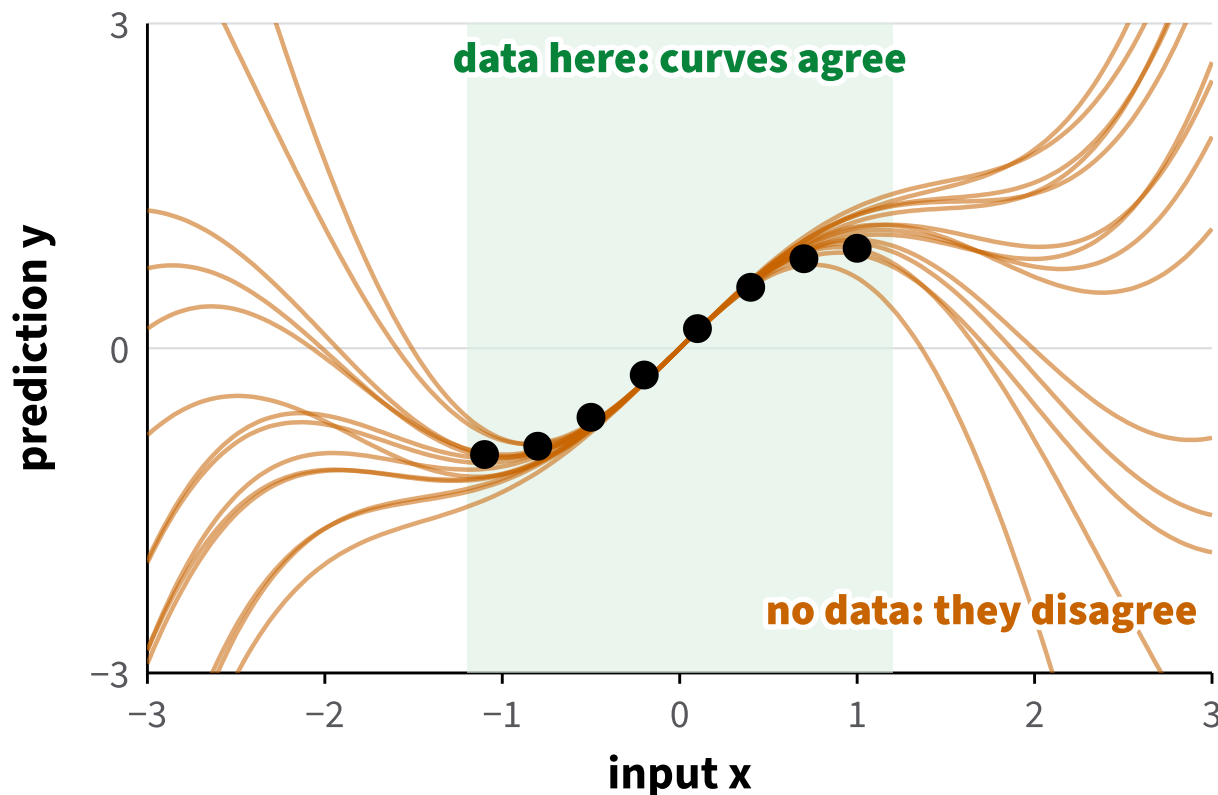
Trouble 3 in three sentences

- BACKGROUND** A neural network is defined by its **weights**. Training keeps one set of weights, but one network cannot tell us **where it is unsure**; for that, the **walker** should visit many good sets of weights.
- TROUBLE** To compare two sets of weights, the **accept test** must score both on **every training example**. With millions of examples, each step is far too slow.
- KEY IDEA** At each step read only a **few random examples**, move in the direction they say improves the fit, add a small **random kick**, and skip the accept test.
- USED FOR** Neural networks that report **how sure** they are about each prediction.

Trouble 3: too much data

Many good networks instead of one: where they **disagree**, the model is **unsure**.

each orange curve: one sampled network (schematic)



Look: each orange curve is one set of weights the **walker** visited. Near the data (black dots) the curves agree; away from the data they spread out. That spread is the **uncertainty**.

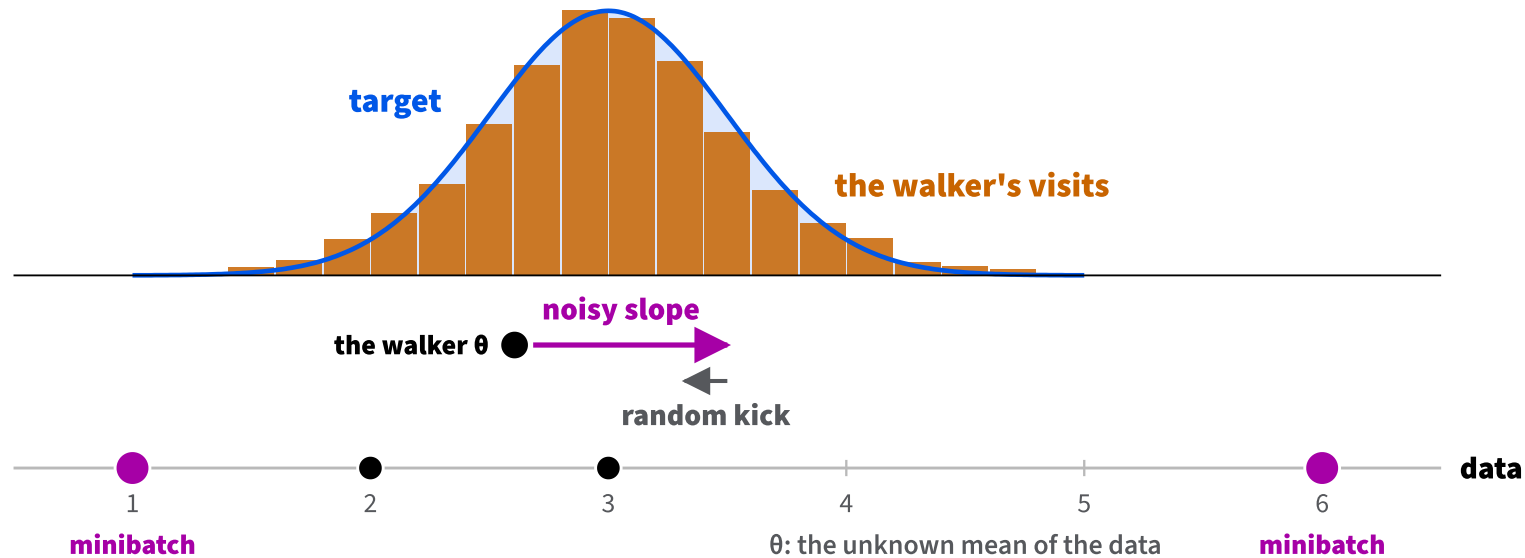
Q: Why not train just one network?

A: One network gives one curve, and it never says “I am not sure”.

Trouble: each **accept test** must score the network on **every training example**. With millions of them, every step is too slow.

Trouble 3: too much data

Fix: read two random examples, step towards them, add a **random kick**; no accept test.



Watch: the unknown θ is the mean of four data points (1, 2, 3, 6). Each step picks two of them (magenta) and moves θ towards their average, the **noisy slope**; then a **random kick** (grey) nudges it. The **visits** (orange) end up spread like the target (blue).

Q: Isn't this just training? A: Without the kick, yes: θ settles near the best value. The kick keeps it moving, so the visits show all good values.

Try: switch the kick off. This is **stochastic-gradient Langevin dynamics** [Welling and Teh, 2011]; skipping the accept test makes the spread slightly too wide.

Outline

Sampling without Z (Lecture 4)

Troubles 1–3: measuring unknowns from data

► **Troubles 4–5: creating pictures and text**

Conclusion and your turn

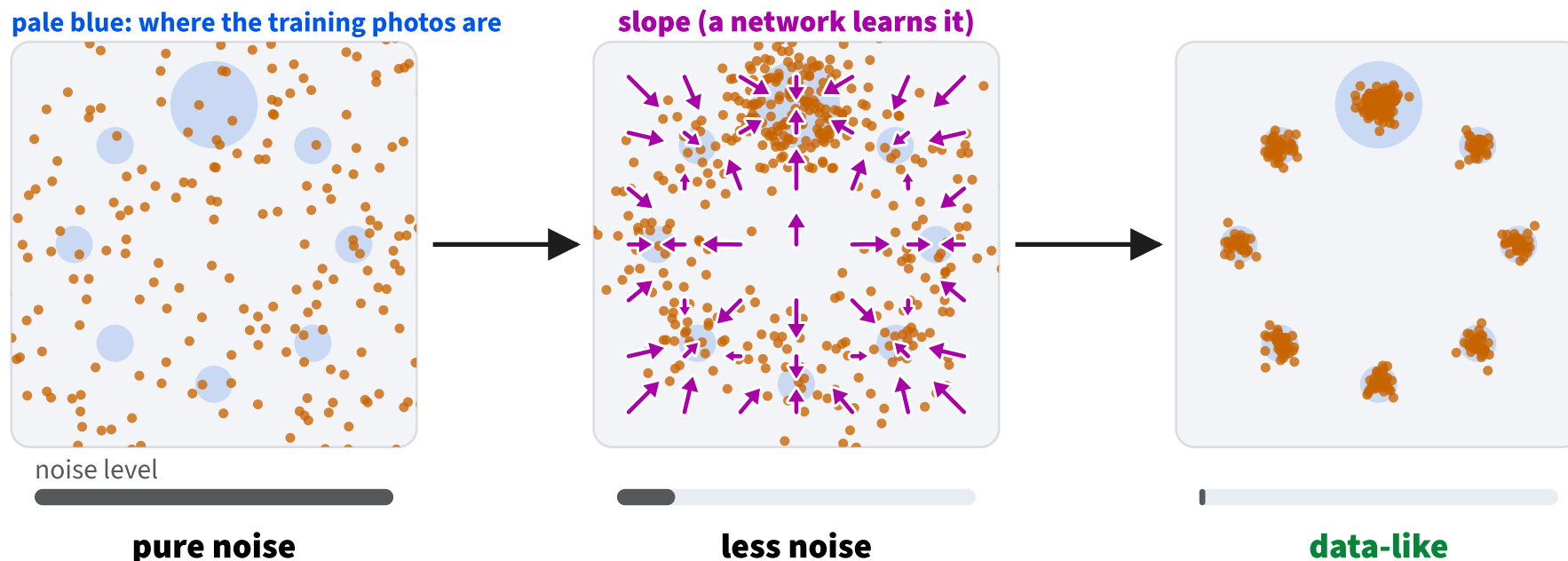
Next: each place the walker visits is a new picture or a new text: it creates.

Trouble 4 in three sentences

- BACKGROUND** For photos there is **no formula** for the **fit score**, only **millions** of example photos.
- KEY IDEA** Train a network to give the **slope** towards “more photo-like”, then **walk from pure noise** while the **noise shrinks**.
- USED FOR** Diffusion image generators, such as Stable Diffusion.

Trouble 4: no formula

Trouble: no formula for $g(\text{photo})$, only examples. **Fix:** learn the slope; walk from noise.



Watch: each dot is one whole picture (one coordinate per pixel; 2 shown). Heavy noise blurs the 8 clusters into one hill, like the **hot walker (p. 14)**; following the slope (**p. 21**) as the **noise shrinks**, the dots settle into all 8, about 40% on the big disk (which holds 44% of the photos).

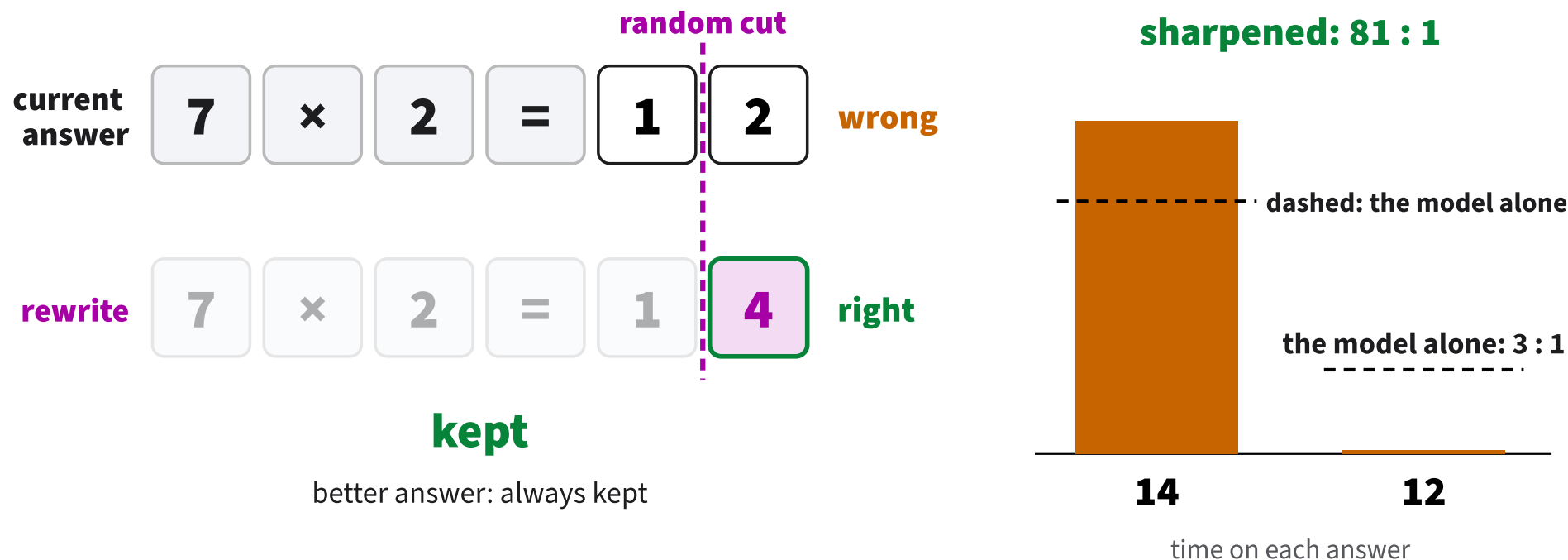
Try: start small: only 14% reach the big disk. Why? This is the idea behind **diffusion image generators** such as Stable Diffusion [Song and Ermon, 2019; Ho et al., 2020].

Trouble 5 in three sentences

- BACKGROUND** A language model gives each answer a probability; the right answer is often the most likely, but not likely enough.
- KEY IDEA** **Sharpen:** **sample** answers with probability raised to a power. The **walker** **rewrites the end** of an answer and keeps it by the **accept test**, so no new Z is needed.
- USED FOR** Better maths and coding answers from the same model, with no extra training (2025).

Trouble 5: a sum over all texts

Trouble: sharpening needs a new Z over all texts. **Fix:** rewrite the end; the accept test decides.



Q: Why not just take the most likely answer? **A:** The likeliest word at each step does not give the likeliest whole answer; finding that needs a search over all texts.

Watch: power 4 turns 3 : 1 into 81 : 1, with no new Z . **Try:** $\alpha = 1$ and $\alpha = 8$. This is **power sampling** [Karan and Du, 2025].

Outline

Sampling without Z (Lecture 4)

Troubles 1–3: measuring unknowns from data

Troubles 4–5: creating pictures and text

► **Conclusion and your turn**

Finally: the five fixes side by side, what is still open, and what you can try yourself.

Conclusion

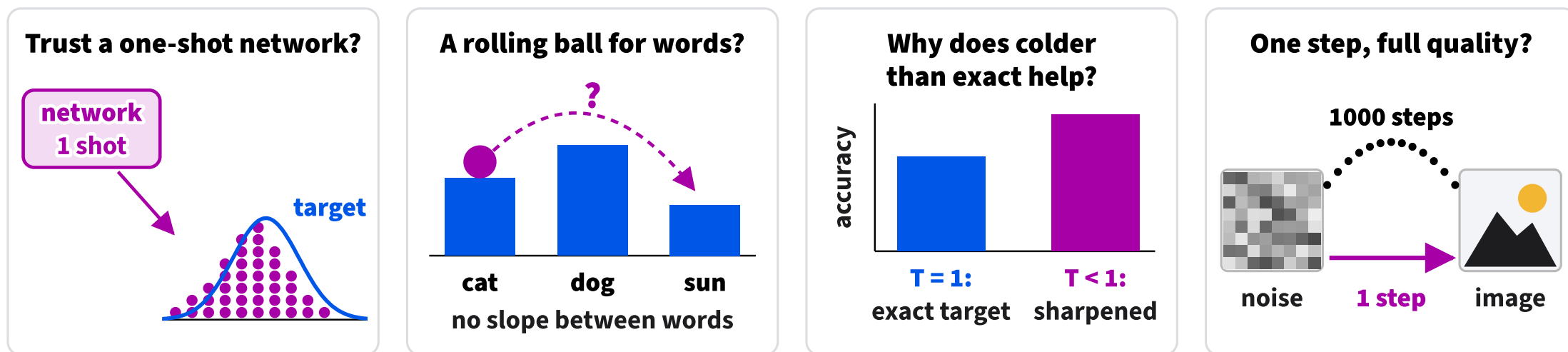
One **walker**, five small changes, and Z is never computed.

	Trouble	Fix	Used for
1	far-apart hills	a hot walker, swaps	colliding black holes
2	many unknowns at once	slide like a marble	COVID-19 lockdowns
3	too much data	minibatch slope + kick	neural-network error bars
4	no formula for g , only examples	learned slope, from noise	image generators
5	sharpening: a new Z over all texts	rewrite the end, test	language-model reasoning

Read: fix 1 adds a hot copy; fixes 2–5 change how the next **candidate** is proposed, and fixes 3–4 also drop the **accept test**. None computes Z : a ratio of **fit scores** cancels it, and a **slope** never sees it.

Conclusion

- ▶ It is possible to **sample without computing Z** : compare two candidates, or follow the **slope**.
- ▶ Each frontier adds a copy, changes the proposal, or drops the **accept test**.
- ▶ We leave four **open problems**:



Think: each would make a good course project; start from Dax et al. (2021) for 1, Wenzel et al. (2020) for 3 and Song et al. (2023) for 4.

Your turn

Predict first, then try: every animation in this lab has a knob.

In these slides

- ▶ the **fit score** of B (p. 6): where do the **walkers** settle?
- ▶ the distance between hills (p. 13): when does the walker cross?
- ▶ the temperature T (p. 14): is hotter always better?
- ▶ the roll length (p. 18) and the **random kick** (p. 21)
- ▶ the noise schedule (p. 24) and the power α (p. 26)

After class: `lab4_explore.py`

1. The walker: do the **visits** follow **fit scores** 1, 2, 5?
2. Far-apart hills: how fast does the wait grow?
3. A **hot walker**: write the **swap** test.
4. Generate from noise: why shrink it slowly?

Course site → Week 4 → Laboratory → Lab files. Each experiment needs one or two lines of Python.

Tip: write your prediction down before you click; the surprises are where the learning is.

Selected References I

Start here: Hastings (1970) explains the walker itself; the others are the methods and today's results.

-  [Abbott, B. P., et al. \(LIGO Scientific Collaboration and Virgo Collaboration\) \(2016\).](#)
Properties of the binary black hole merger GW150914.
Physical Review Letters, 116(24):241102.
-  [Dax, M., Green, S. R., Gair, J., Macke, J. H., Buonanno, A., and Schölkopf, B. \(2021\).](#)
Real-time gravitational wave science with neural posterior estimation.
Physical Review Letters, 127(24):241103.
-  [Duane, S., Kennedy, A. D., Pendleton, B. J., and Roweth, D. \(1987\).](#)
Hybrid Monte Carlo.
Physics Letters B, 195(2):216–222.
-  [Event Horizon Telescope Collaboration \(2019\).](#)
First M87 Event Horizon Telescope results. VI. The shadow and mass of the central black hole.
The Astrophysical Journal Letters, 875(1):L6.
-  [Flaxman, S., Mishra, S., Gandy, A., et al. \(2020\).](#)
Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe.
Nature, 584:257–261.
-  [Geyer, C. J. \(1991\).](#)
Markov chain Monte Carlo maximum likelihood.
In Computing Science and Statistics: Proceedings of the 23rd Symposium on the Interface, pages 156–163.

Selected References II

Good entry points: Song and Ermon (2019) for image generation; Karan and Du (2025) for language models.



Hastings, W. K. (1970).

Monte Carlo sampling methods using Markov chains and their applications.

Biometrika, 57(1):97–109.



Ho, J., Jain, A., and Abbeel, P. (2020).

Denoising diffusion probabilistic models.

In Advances in Neural Information Processing Systems 33, pages 6840–6851.



Karan, A. and Du, Y. (2025).

Reasoning with sampling: your base model is smarter than you think.

arXiv:2510.14901.



Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., and Teller, E. (1953).

Equation of state calculations by fast computing machines.

The Journal of Chemical Physics, 21(6):1087–1092.



Song, Y. and Ermon, S. (2019).

Generative modeling by estimating gradients of the data distribution.

In Advances in Neural Information Processing Systems 32.

Selected References III

A short read: Welling and Teh (2011), eight pages on how sampling meets minibatch training.



[Song, Y., Dhariwal, P., Chen, M., and Sutskever, I. \(2023\).](#)

Consistency models.

In Proceedings of the 40th International Conference on Machine Learning, PMLR 202:32211–32252.



[Swendsen, R. H. and Wang, J.-S. \(1986\).](#)

Replica Monte Carlo simulation of spin-glasses.

Physical Review Letters, 57(21):2607–2609.



[Veitch, J., Raymond, V., Farr, B., et al. \(2015\).](#)

Parameter estimation for compact binaries with ground-based gravitational-wave observations using the LALInference software library.

Physical Review D, 91(4):042003.



[Welling, M. and Teh, Y. W. \(2011\).](#)

Bayesian learning via stochastic gradient Langevin dynamics.

In Proceedings of the 28th International Conference on Machine Learning, pages 681–688.



[Wenzel, F., Roth, K., Veeling, B. S., Świątkowski, J., Tran, L., Mandt, S., Snoek, J., Salimans, T., Jenatton, R., and Nowozin, S. \(2020\).](#)

How good is the Bayes posterior in deep neural networks really?

In Proceedings of the 37th International Conference on Machine Learning, PMLR 119:10248–10259.

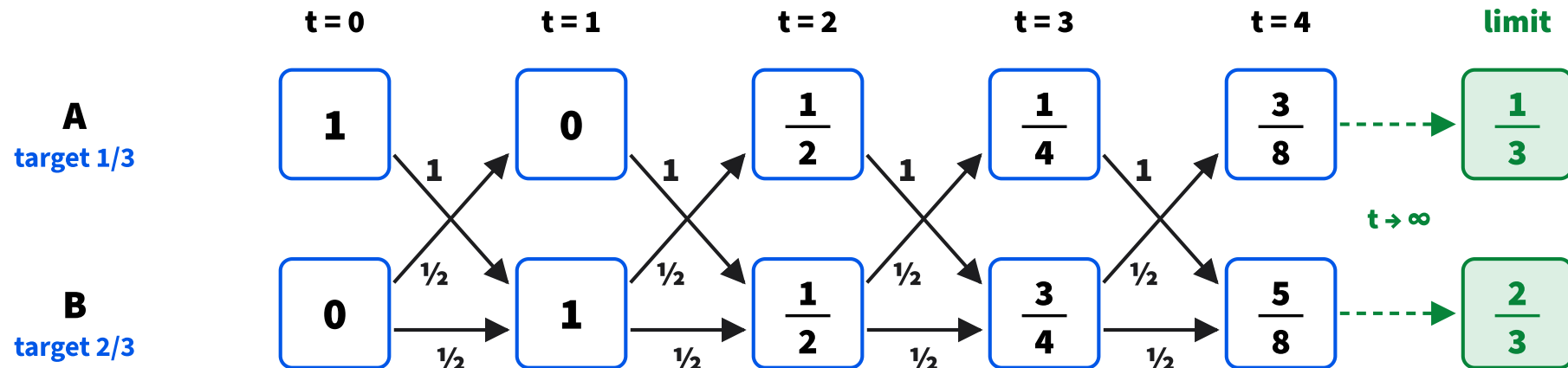
Thank you!

Next: Lab 5 (Saturday 10 October), the expectation–maximization algorithm at the frontier

Try the walker yourself: the ten-line Python version is on the backup slides that follow.

Backup: the walker of Lecture 4

Why the **walkers** settle at one third and two thirds: the flows balance.



Look: each box is the share of walkers at step t : the sum of (previous box $\times$ arrow). The shares stop changing when the flow $A \rightarrow B$, $\frac{1}{3} \cdot 1$, equals the flow $B \rightarrow A$, $\frac{2}{3} \cdot \frac{1}{2}$: detailed balance (Lecture 4, Theorem 9).

Backup: the walker of Lecture 4

The whole **walker** is a few lines of Python, and it never uses Z .

```
import random
random.seed(0)
g = {"A": 1, "B": 2}                                # fit scores; Z is never computed
x, visits = "A", {"A": 0, "B": 0}
for t in range(100_000):
    y = "B" if x == "A" else "A"                    # propose the other picture
    if random.random() < min(1, g[y] / g[x]):        # the accept test: better fit always, worse g(new)/g(old)
        x = y
    visits[x] += 1
print({s: v / 100_000 for s, v in visits.items()})
{'A': 0.33392, 'B': 0.66608}
```

Try: change the **fit scores** 1 and 2 to any two numbers; the shares follow their ratio, and Z is still never computed. More in `lab4_explore.py`.

Backup: Trouble 1

The **swap** of parallel tempering is again a comparison.

A copy at temperature T walks on $g(x)^{1/T}$. Two copies at $T_1 < T_2$ propose to swap their states x_1, x_2 ; the **accept test** compares the swapped pair with the current pair:

$$\frac{g(x_2)^{1/T_1} \cdot g(x_1)^{1/T_2}}{g(x_1)^{1/T_1} \cdot g(x_2)^{1/T_2}} = \left(\frac{g(x_2)}{g(x_1)} \right)^{\frac{1}{T_1} - \frac{1}{T_2}}.$$

Accept with $\min\{1, \text{this ratio}\}$. In the animation $T_1 = 1$ and $T_2 = 10$, so the power is $\frac{9}{10}$.

Notice: only the ratio $g(x_2)/g(x_1)$ appears, and the swap keeps the **walker's visits** correct. Like every step, it never needs Z .

Backup: Trouble 2

One leapfrog step, by hand (a roll chains many of them).

Height $U(x) = -\ln g(x) = \frac{1}{2} \cdot x^2$, **slope** $U'(x) = x$. Start at $x = 1$ with **momentum** $v = 1$, step 0.5; energy $H = U(x) + \frac{1}{2} \cdot v^2 = 1$.

$$v = 1 - 0.25 \cdot 1 = 0.75$$

$$x = 1 + 0.5 \cdot 0.75 = 1.375$$

$$v = 0.75 - 0.25 \cdot 1.375 = 0.406$$

$$H' = \frac{1}{2} \cdot 1.375^2 + \frac{1}{2} \cdot 0.406^2 = 1.028$$

$$\min\{1, e^{-(1.028-1)}\} = 0.97$$

half kick: speed $- 0.25 \cdot$ slope

drift: position $+ 0.5 \cdot$ speed

half kick

energy almost kept

the **accept test**, with **fit score** e^{-H}

Notice: in this one step the energy moves only from 1 to 1.028, so the accept test keeps it 97% of the time; a roll is many such steps.